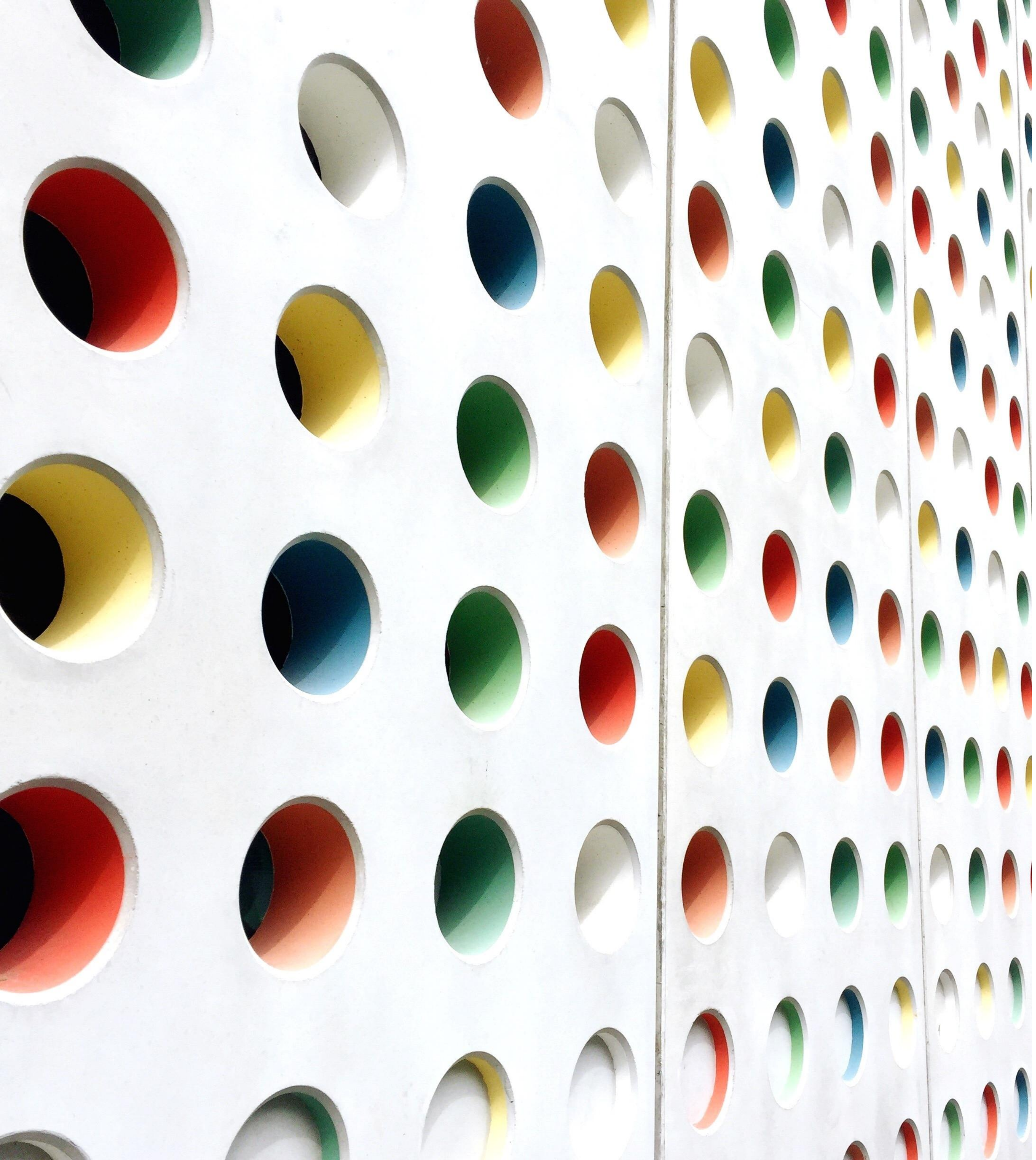

Semantic Web

Semantic Web Languages

BY

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Machine Processable Semantics

- Machine is any device that uses energy to perform some activity
- Any piece of equipment made for a particular purpose
- Purpose is a result, end, aim, or goal of an action intentionally undertaken
- An agent's intention in performing an action is his or her specific purpose
- Semantics is the study of meaning a range of ideas
- Database is a collection of data" and "the term data means groups of information

In a relational database, everything is represented in a table, and a row has a key, and a column has a name. With this, even with a very simple machine, one can find the phone number of Mr. X if X is the value of the name column and phone number is the heading of another column. Unfortunately, with an average Web page, this is far more difficult. Hidden in various HTML tags there is a name (a random alphanumerical string similar to many others) and somewhere else a phone number (a set of integers including some special characters). A browser is required to render the information and a human reader to understand the information based on the layout of the website.

Essence of Semantic Web technology

Semantic technology adds tags to semi-structured information as database technology adds column headings to tabular information. Let us use a small example:

<person>

<name>Sir Tim</name>

<phone number>01-444444</phone number>

</person>

These annotations allow a computer "to understand" that Sir Tim is a name of a person and 01-444444 is his telephone number.

Programs and other computational resources can be described through semantic annotations. One needs two things to define the semantics of information: a language such as <X>Y</X> to define the meaning of Y, and terms such as X to denote this meaning.

Form

The focus is on a small number of languages, on those that provide insights into the overall design issues associated with logical languages and those that have been applied in a Semantic Web context.

A number of languages will be then examined that are used to express the meaning of data on the Semantic Web.

Logic

- Prepositional Logic
- First-Order Predicate Logic
- Second-Order Predicate Logic
- Computational logic
- Horn Logic

Semantic Web Languages

HTML provides several ways to express the semantics of data. An obvious one is the META tag :

<META name = “Author” lang= “fr” content = “Arnaud Le Hors”>

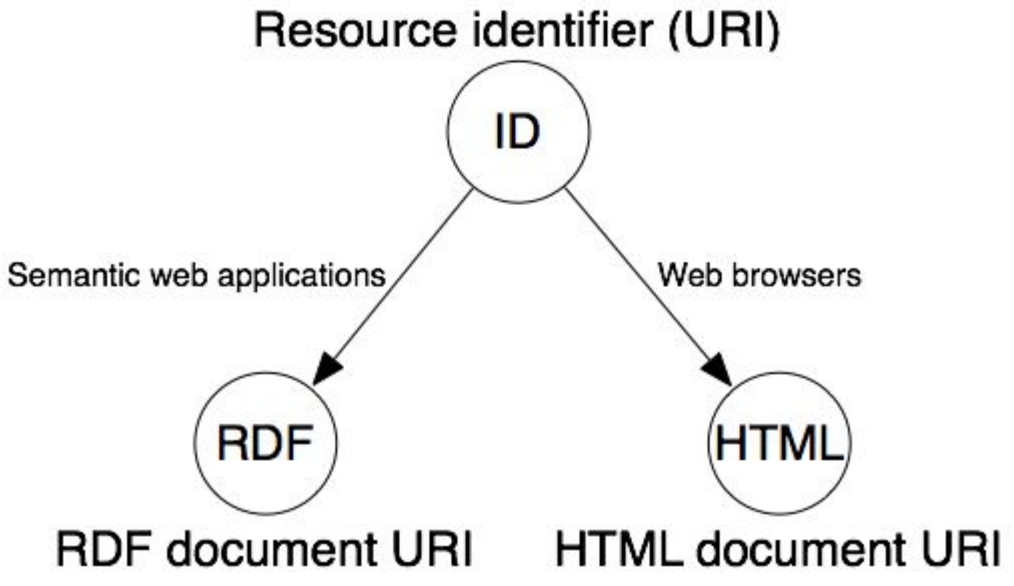
Before the wider usage of RDF, systems such as Ontobroker used the attribute of the anchor tag to encode semantic information. It is also possible to interpret the semantics of HTML documents indirectly.

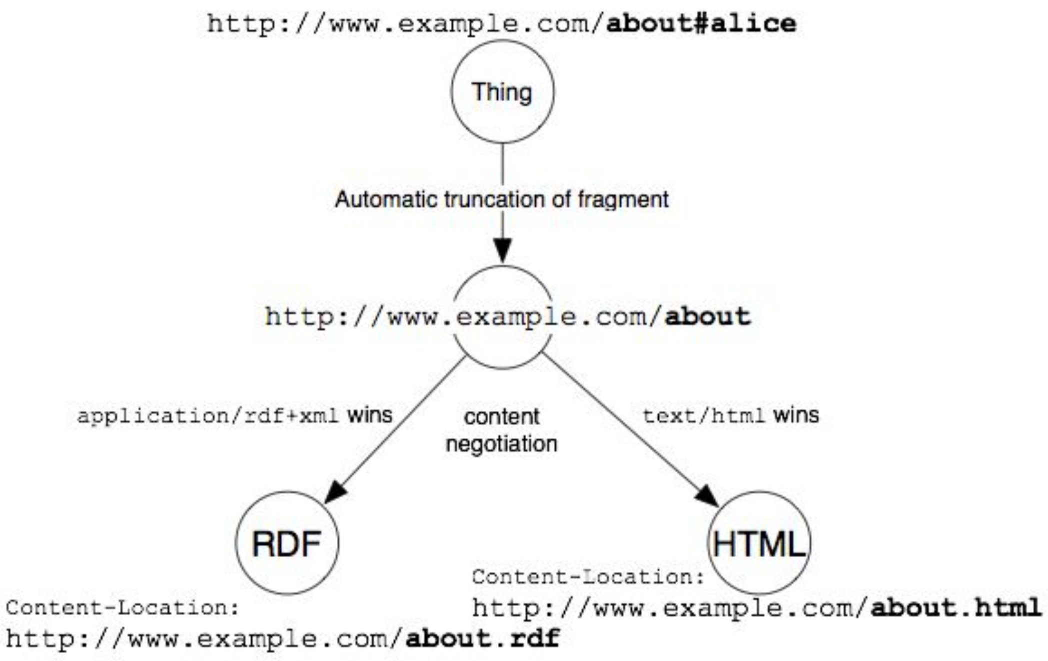
Within efforts to stretch the use of HTML to include meaning, the term semantic HTML was created

What are XML, RDF

1. Xml: The Extensible Markup Language (XML) has been developed as a generic way to structure documents on the Web. It generalizes HTML by allowing user-defined tags. This flexibility of XML, however, reduces the possibilities for the type of semantic interpretation that was possible with the predefined tags of HTML.

2. RDF: Resource Description Framework (RDF) is a simple data model for semantically describing resources on the Web. Binary properties interlink terms forming a directed graph. These terms as well as the properties are described using URIs. Since a property can be a URI, it can again be used as a term interlinked to another property.





Conti... [RDF]

1. Unlike most logical languages or databases, it is not possible to distinguish the language or schema from statements in the language or schema. For example, in the statement

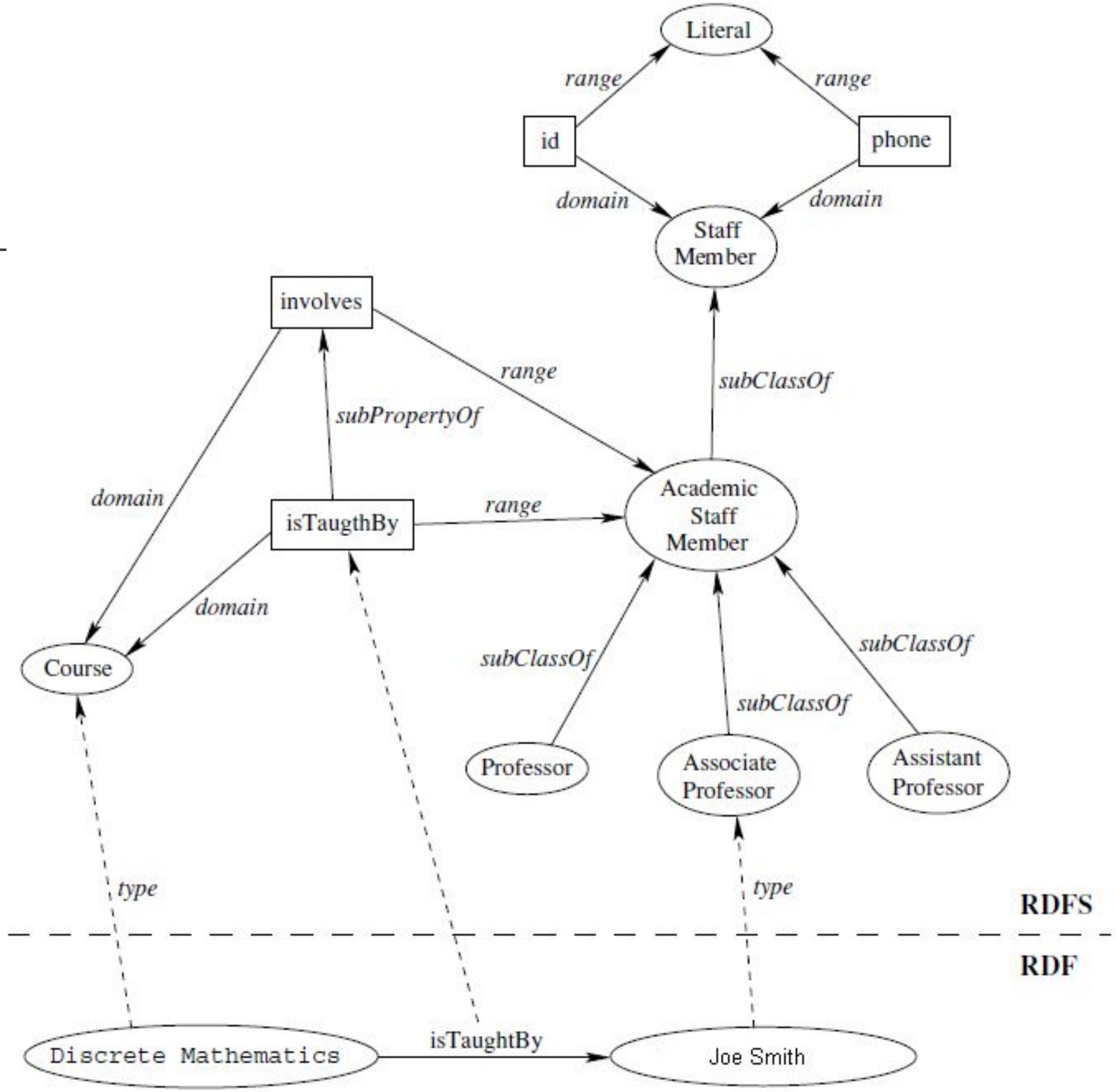
<rdf:type, rdf:type, rdf:Property>

it is stated that *type* is of *type property*.

2. Unlike conventional hypertext, in RDF, URIs can refer to any identifiable thing (e.g., a person, vehicle, business, or event). This very flexible data model is obviously suitable in the context of a free and open Web.

RDFS

- RDF schema (RDFS) uses basic RDF statements and defines a simple ontology language.
- It defines entities such as rdfs:class, rdfs:subclass, rdfs:subproperty, rdfs:domain, and rdfs:range, enabling one to model classes, properties with domain and range restrictions, and hierarchies of classes and properties.
- RDFS is a specific RDF vocabulary for this purpose and is simply RDF plus some more definitions (statements) in RDF.



OWL

- Web Ontology Language OWL extends this vocabulary to a full-fledged spectrum of Descriptions Logics defined in RDF
  - OWL Lite
  - OWL DL
  - OWL Full
- Mechanisms are provided to define properties to be inverse, transitive, symmetric, or functional.
- Properties can be used to define the membership of instances for classes or hierarchies of classes and of properties.
- OWL Lite is already quite an expressive Description Logic which makes the development of efficient implementations for large data sets quite challenging and difficult as implementing OWL DL.
- However, neither of these languages can make use of full RDF, some valid RDF statements are not valid in Lite or DL. This is because logic languages such as Descriptions Logics exclude meta statements, that is, statements over statements.
- For RDF and RDFS, this was not a problem since neither language provided mechanisms to define complex logical definitions.
- Lite and DL define a vocabulary in RDF and restrict the usage of RDF.
- OWL Full drops these restrictions. OWL Full provides the vocabulary of OWLDL, that is, an expressive Description Logic, and allows for any valid RDF statement. For example, in OWL Full, a class can be treated simultaneously as a set of individuals and as an individual. Therefore, OWL Full is beyond the expressive scope of Description Logic and minimally requires a theorem prover type of inference such as first order logic (i.e., is semi-decidable).

OWL Type Features

1. OWL Lite was designed for easy implementation and to provide users with a functional subset that will get them started in the use of OWL.

2. OWL DL (where DL stands for "Description Logic") was designed to support the existing Description Logic business segment and to provide a language subset that has desirable computational properties for reasoning systems.

3. The complete OWL language (called OWL Full to distinguish it from the subsets) relaxes some of the constraints on OWL DL so as to make available features which may be of use to many database and knowledge representation systems, but which violate the constraints of Description Logic reasoners.

Note: RDF documents will generally be in OWL Full

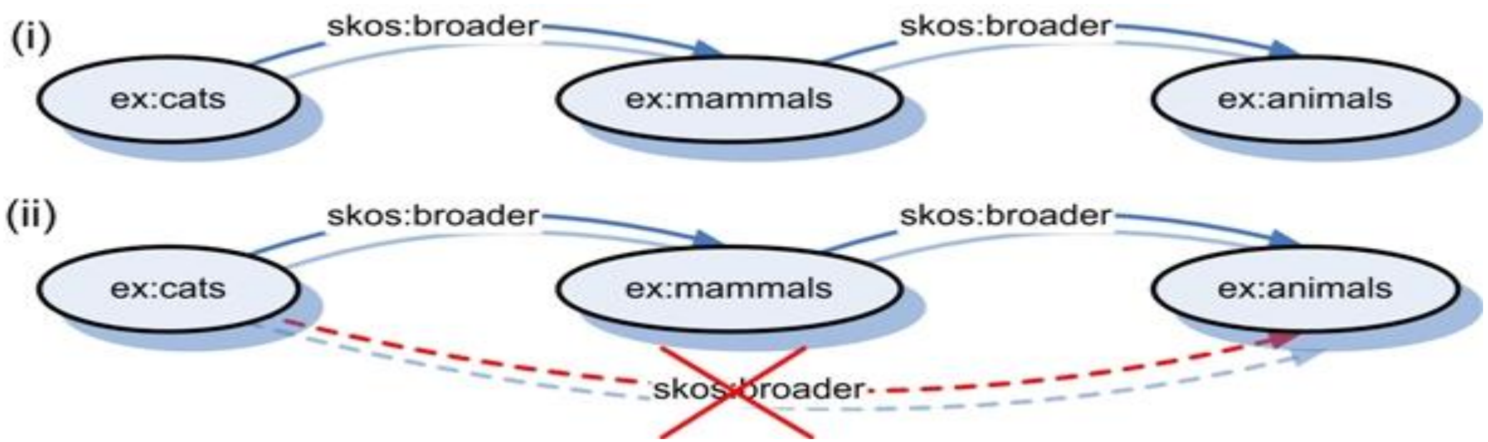
Restrictions

- OWL Full and OWL DL support the same set of OWL language constructs. Their difference lies in restrictions on the use of some of those features and on the use of RDF features.
- OWL Full allows free mixing of OWL with RDF Schema and, like RDF Schema, does not enforce a strict separation of classes, properties, individuals and data values.
- OWL DL puts constraints on the mixing with RDF and requires disjointness of classes, properties, individuals and data values. The main reason for having the OWL DL sublanguage is that tool builders have developed powerful reasoning systems which support ontologies constrained by the restrictions required for OWL DL.

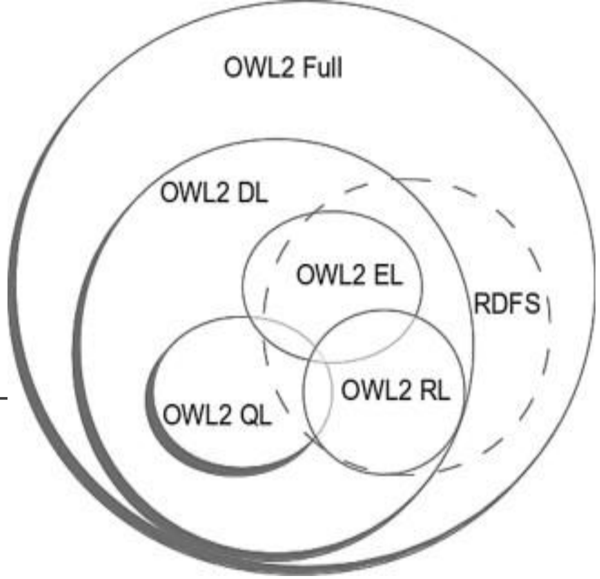
Note: RDF users upgrading to OWL should be aware that OWL Lite is *not* simply an extension of RDF Schema. OWL Lite is a light version of OWL DL and puts constraints on the use of the RDF vocabulary (*e.g.*, disjointness of classes, properties, etc.). OWL Full is designed for maximal RDF compatibility and is therefore the natural place to start for RDF users.

SKOS -- OWLFull

- OWLFull can be used as a basis to find useful restrictions (OWL DL is an example of such a restriction) and generate useful languages such as the Simple Knowledge Organization System (SKOS).
- SKOS is a data model for knowledge organization systems that uses keywords to describe resources. SKOS is defined as an OWL Full ontology, that is, it uses a sub-vocabulary of OWL Full to define a vocabulary for simple resource descriptions based on controlled structured vocabularies.



OWL2<— OWL DL

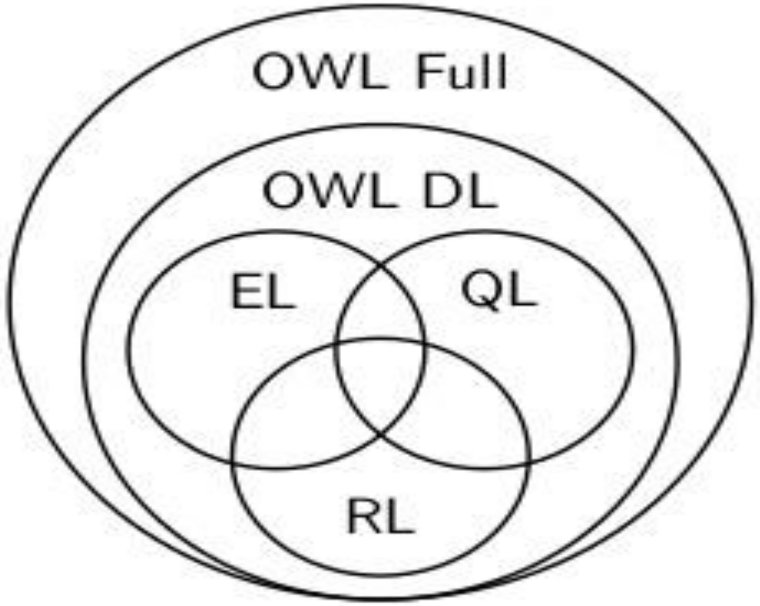


- OWL Lite had been defined as an over expressive Description Logic. This hampered the implementation of Lite reasoning based on existing semantic repository technologies and made the layering of rules on top of the language unfeasible.
- Specifically, there was too big a gap between RDFS and OWL Lite. In consequence, three new sub-languages were defined.
  - OWL2EL: It provides polynomial time algorithms for all the standard reasoning tasks of description logic.
  - OWL2QL: It enables efficient query answering over large instance populations.
  - OWL2RL: It restricts the expressiveness with respect to extensibility toward rule languages. OWL2RL seamlessly links with rule-based presentations of RDFS and extensions to simple rule languages. This is currently the route that most industrial semantic repository developers follow and will probably define together with OWL2QL the most important Semantic Web representation languages from a technological point of view.

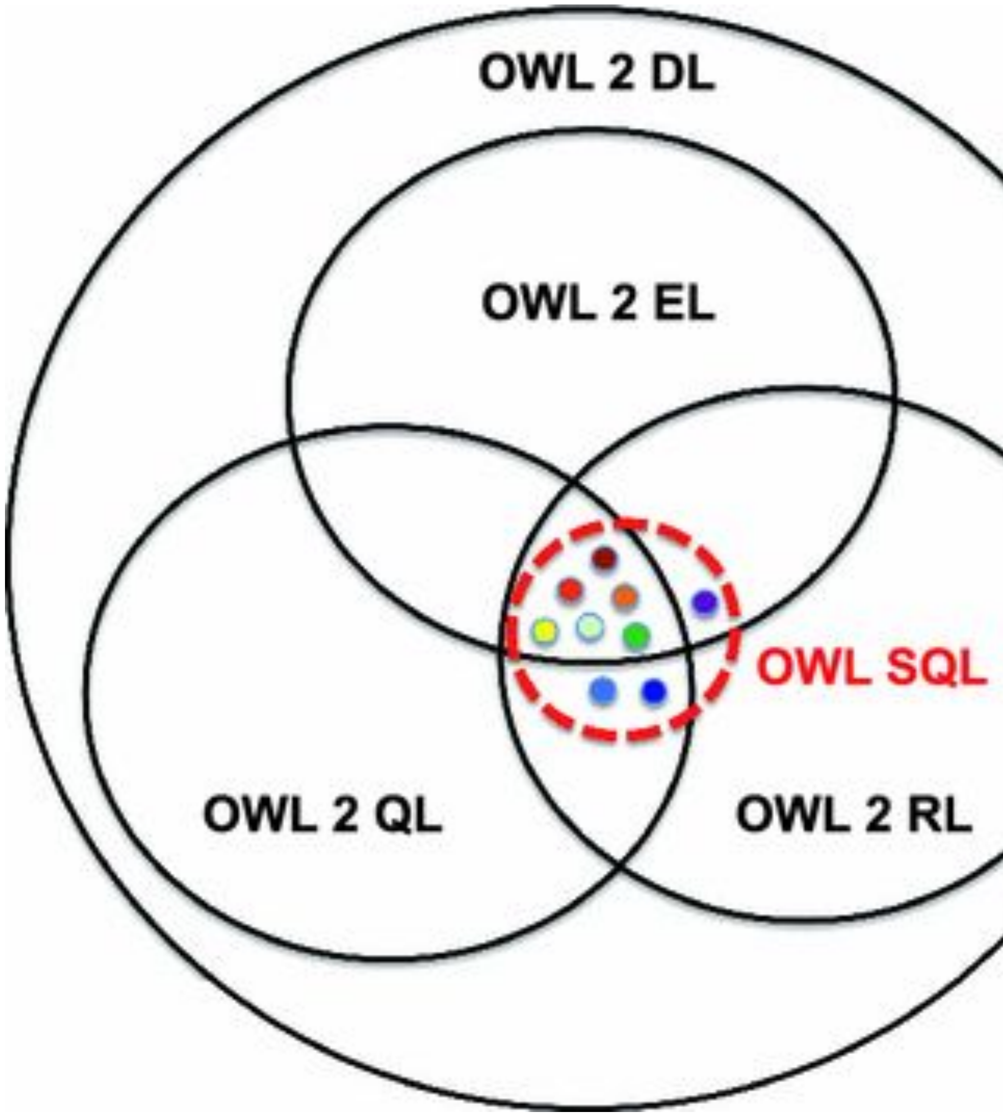
OWL 2 profiles

OWL 2 DL can be seen as a "group" of sublanguages called *profiles*:

- OWL 2 EL, suitable for big and relatively simple taxonomies
- OWL 2 QL, suitable for conjunctive queries on many instances
- OWL 2 RL, a sort of "compromise" between expressivity and scalability inspired by rule-based reasoning



- Recent proposal: OWL-LD (OWL for the Linked Data) → `http://semanticweb.org/OWLLD/`



		EL	QL	RL
	subClass	X	X	X
	subProp	X	X	X
	domain	X	X	X
	range	X	X	X
	eqClass	X	X	X
	eqProp	X	X	X
	inverseProp		X	X
	symProp		X	X
	transProp	X		X

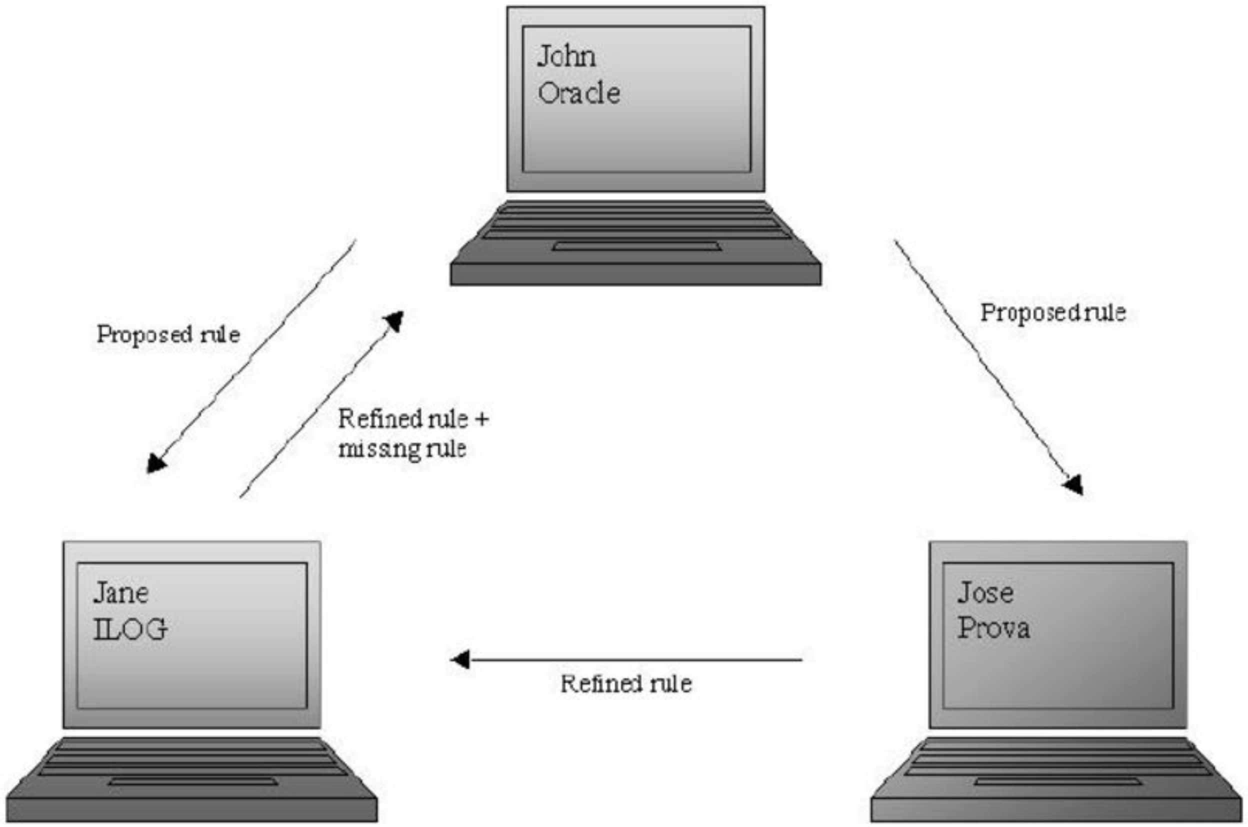
RIF/SWRL

- Rule Interchange Format (RIF) complements OWL with a language framework centered on the rule paradigm/ it is the part of the infrastructure of semantic web
- Like OWL, it does not come as a single language but as several sub-languages.
- RIF includes three dialects, a Core dialect which is extended into a Basic Logic Dialect (BLD) and Production Rule Dialect (PRD)
- The framework incorporates RIF-BLD, which defines a simple logic-oriented rule language;
  - RIF-PRD: which captures most of the aspects of production rule systems
  - RIF-Core: which is the intersection of both these languages.

This split is because the W3C working group had to cover two very different paradigms which are only similar at the surface level:

1. Rules based on a declarative interpretation of logic
2. Rules that model event–action systems based on the production rule paradigm.

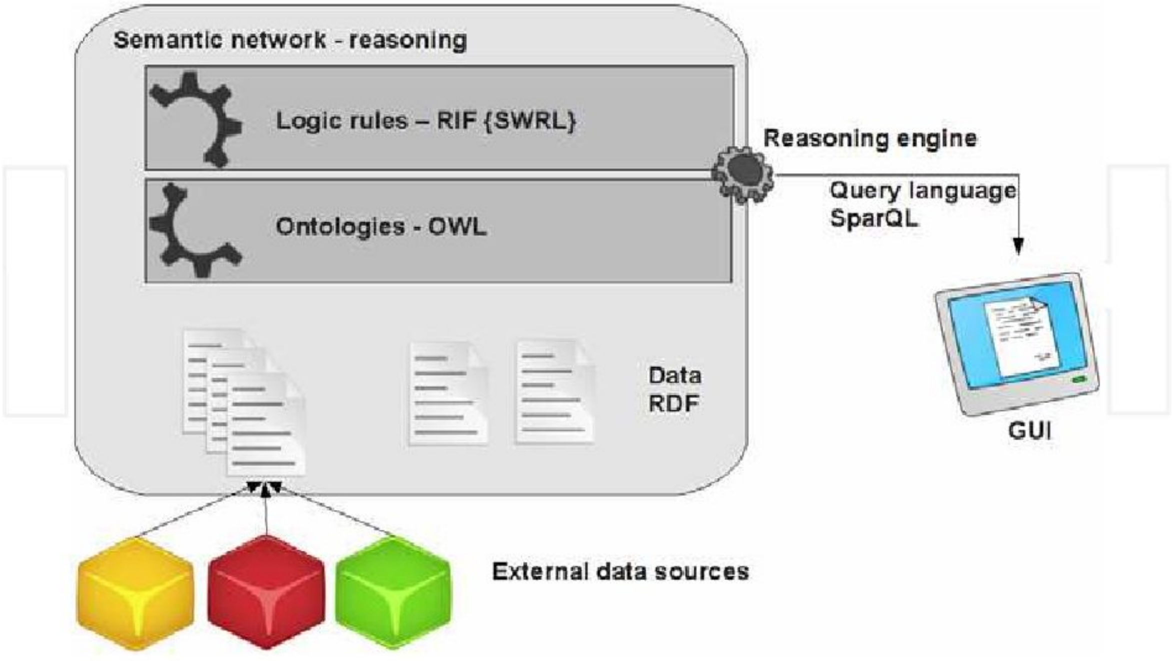
RIF



John
Oracle
Proposed rule
Refined rule +
missing rule
Proposed rule
Jane
ILOG
Refined rule
Jose
Prova

Rule Interchange Format (RIF)

- Goals
  - Define a framework for rule languages for the Semantic Web
    - If *<condition>* then *<conclusion>*
  - Define a standard format/syntax for interchanging rules
- Several sublanguages defined so far
  - different expressivity & complexity
  - ***RIF BLD*** (Basic Logic Dialect)
    - Rule condition/conclusion are *monotonic* (like in OWL/RDF)
  - ***RIF PRD*** (Production Rule dialect)
    - Condition/conclusion are *non-monotonic* (retraction)



SPARQL

RDF is a data model; it also requires a query language. As SQL is a means to express queries over relation databases, SPARQL is a query language for the graph-based data model of RDF. SPARQL has been developed without considering RDFS, OWL, and RIF.

http://Site1.com/RDF

```
:a1 :Title "Web 2.0"
:a1 :Author "Hacker B."
:a1 :Year  2007
:a1 :Publisher "Springer"
:a2 :Title "Web 3.0"
:a2 :Author "Smith B."
```

http://Site2.com/RDF

```
:4 :Title "Semantic Web"
:4 :Author "Tom Lara"
:4 :PubYear  2005
:5 :Title "Web services"
:5 :Author "Bob Hacker"
```

Query:

```
PREFIX S1: <http://site1.com/rdf>
PREFIX S2: <http://site1.com/rdf>
SELECT ? ArticleTitle
FROM <http://site1.com/rdf>
FROM <http://site2.com/rdf>
WHERE {
 {{?X S1:Title ?ArticleTitle}UNION
 {?X S2:Title ?ArticleTitle}}
 {?X S1:Author ?X1} UNION {?X S2:Author ?X1}
 {?X S1:PubYear ?X2} UNION {?X S2:Year ?X2}
 FILTER regex(?X1, "^Hacker")
 FILTER (?X2 > 2000)}
```

Results:

ArticleTitle
Web 2.0

Figure 1. An example of a SPARQL query.

Microformats – RDFa, GRDDL

<person>

<name>Sir Tim</name>

<phone number>01-444444</phone number>

</person>

A way to define a concept person and properties such as name and phone number has been developed, but there is yet no mechanism to express that Sir Tim is the name of a person. Grounding or connecting metadata with documents on the Web is supported by a set of languages.

- Microformats are predefined formats to add meta information to elements in HTML and XML.
  - A well-known microformat is hCard, which can be used for representing people, companies, organizations, and places.
  - vCard, a file format standard for electronic business cards. These formats not only provide a language structure to present information but additionally provide domain-specific terminologies (controlled vocabularies) for this purpose.

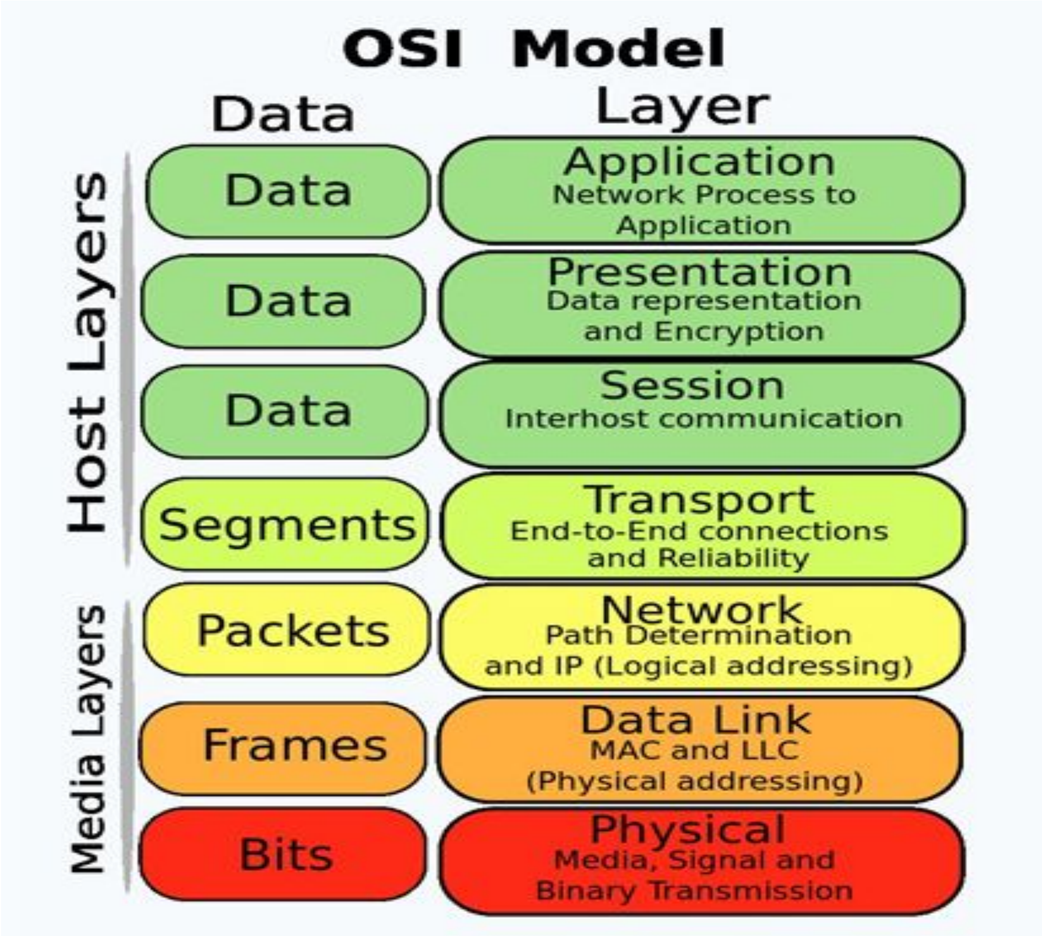
  They directly interweave structure and content.

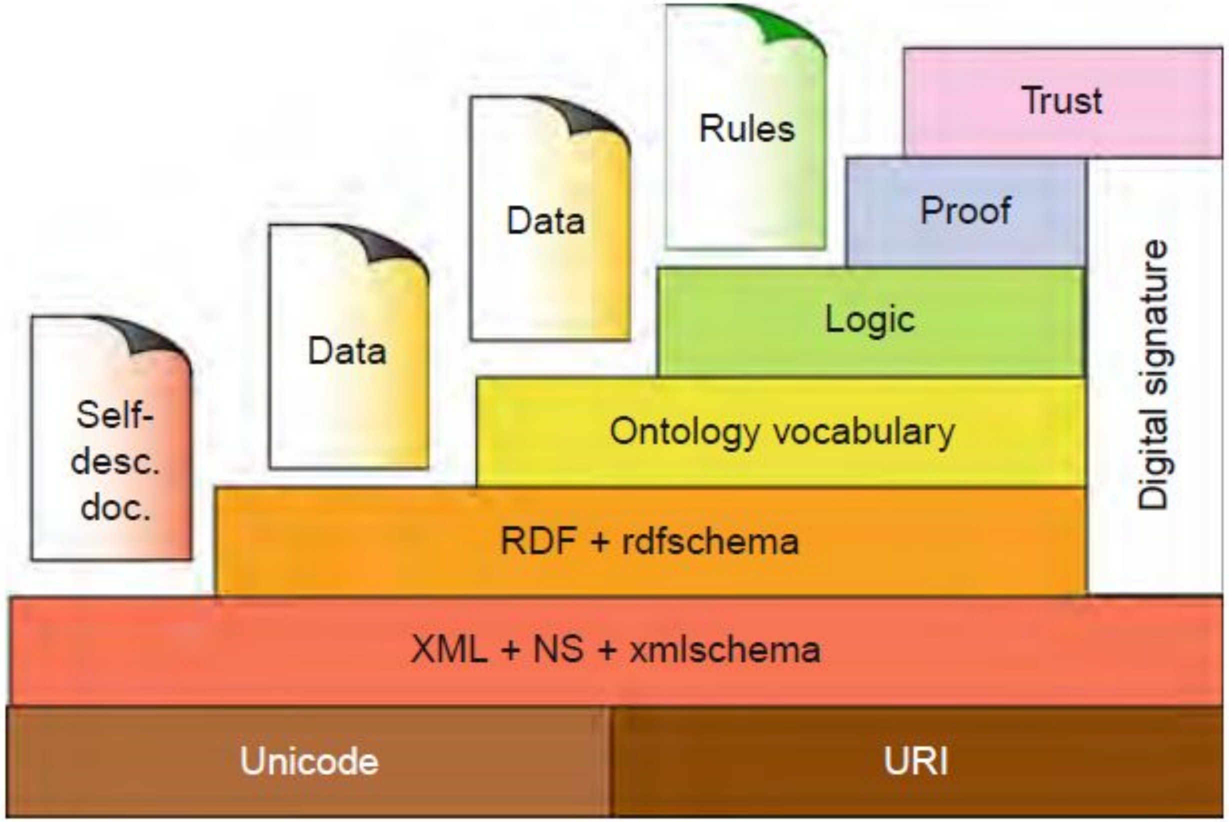
  - RDFa provides a set of XHTML attributes to include RDF metadata directly into HTML and XML documents. In contrast to Microformats, RDFa does not predefine domain-specific terminologies.
  - GRDDL has been developed as a mechanism for Gleaning Resource Descriptions from Dialects of Languages to derive RDFa definitions from Microformats helping to integrate information from heterogeneous sources.

The Tower of Babel

The Open Systems Interconnection model (OSI model) is a product of the Open Systems Interconnection effort at the International Organization for Standardization. It is a way of sub-dividing a communications system into smaller parts called layers. A layer is a collection of conceptually similar functions that provide services to the layer above it and receives services from the layer below

Traditional Web Layers vs Semantic Web Layer Cake





Layering functionalities

This type of layering has two major functions:

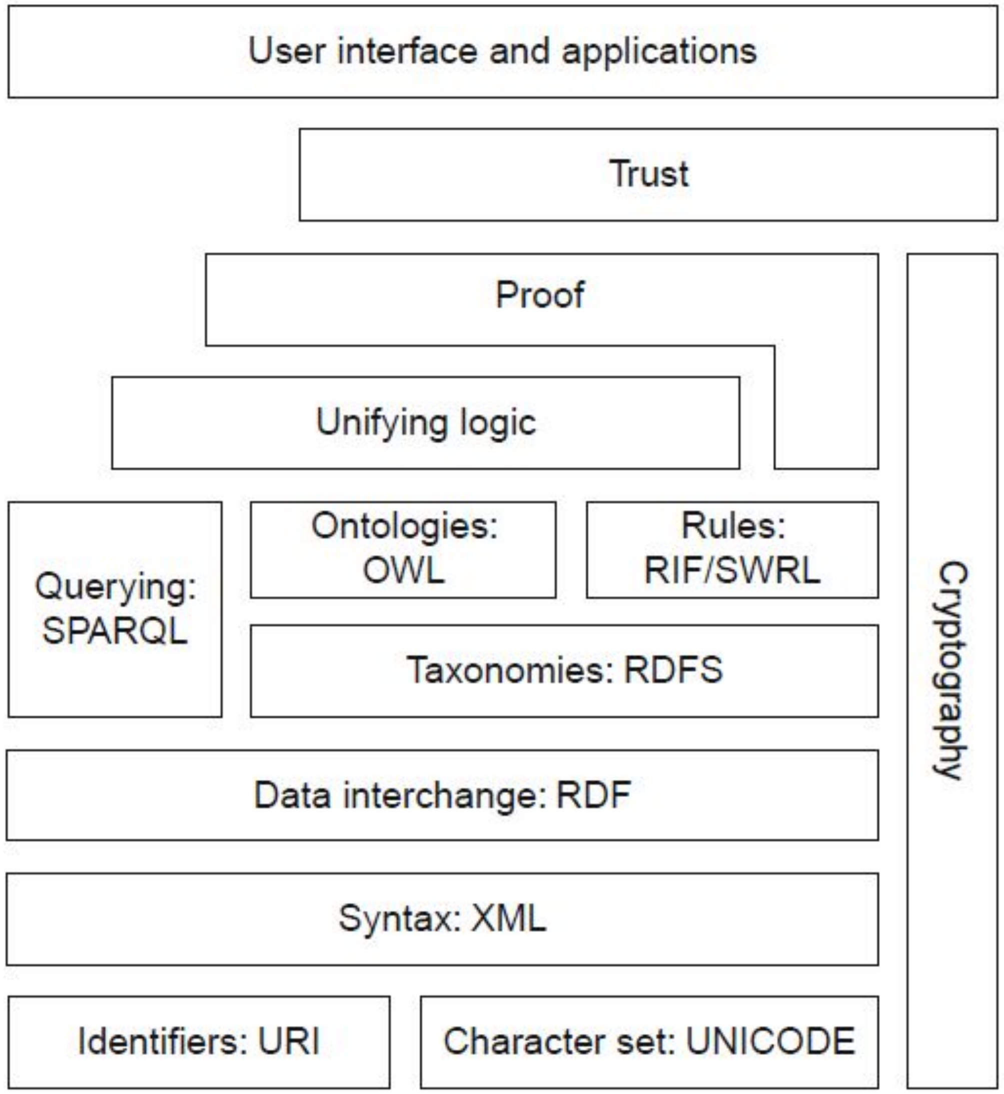
1. Preventing an upper layer from re-implementing functionality provided by a layer below.

2. Allowing an application that only understands a lower layer to at least interpret portions of definitions at a higher layer.

Objection and Rules

- The design should be such that agents fully aware of a layer should be able to take at least partial advantage of information at higher levels. For example, an agent aware only of the RDF and RDF Schema semantics might interpret knowledge written in OWL partly, by disregarding those elements that go beyond RDF and RDF Schema. Of course, there is no requirement for all tools to provide this functionality; the point is that this option should be enabled
- The rationale for having a separate OWL class construct lies in the restrictions on OWL DL (and thus also on OWL Lite), which imply that not all RDFS classes are legal OWL DL classes
- Downward compatibility. Agents fully aware of a layer should also be able to interpret and use information written at lower levels. For example, agents aware of the semantics of OWL can take full advantage of information written in RDF and RDF Schema.

Semantic Web Cake Layer 2



Difference between rule and Description Logic languages

- Interpret integrity constraints, such as the domain and range restrictions of properties. When the value of a property is found and it is not known that it is a member of its range, it is assumed that there must be a mistake. The violation of a constraint is indicated over the range of the property.
- An RDF vocabulary might describe limitations on the types of values that are appropriate for some property, or on the classes to which it makes sense to ascribe such properties. The RDF Vocabulary Description language provides a mechanism for describing this information but does not say whether or how an application should use it. For example, while an RDF vocabulary can assert that an author property is used to indicate resources that are instances of the class Person, it does not say whether or how an application should act in processing that range information. Different applications will use this information in different ways. For example, data checking tools might use this to help discover errors in some data set, an interactive editor might suggest appropriate values, and a reasoning application might use it to infer additional information from instance data. RDF vocabularies can describe relationships.

Already from this statement, you can trace the branching of OWL and RIF.

Difference between rule and Description Logic languages

- RIF has the fundamental problem of covering rule languages based on very different paradigms incorporating either a declarative or an operational flavor. It is of no surprise that RIF is not a single language but, within its first version, provides three languages.
- OWL now provides at least six different dialects. Thus in total, one has more than ten Semantic Web languages, and RIF additionally contains a framework for defining more.
- This language fragmentation is quite dangerous as it may significantly hamper information interoperability between Semantic Web applications and also significantly increase the effort to implement them.

Difference between rule and Description Logic languages

- Examine the layering of SKOS. SKOS uses RDF and OWL. Tt should be assumed that SKOS is layered on top of OWL. However, it is simpler than OWL.
- OWL is supposed to provide a language for defining ontologies, and SKOS is a way to define simple "taxonomies/ classfications." Therefore, it does not extend OWL but rather defines a small extension of a heavily constrained restriction of OWL.
- One would simply expect to define OWL as an extension of SKOS. Not to mention that SKOS is uncertain regarding whether its restricted version of OWL is interpreted as OWL DL or OWL Full.

THE END